

AI-Powered Tools for Supporting Software Engineers Across the Software Lifecycle

Software engineering is a complex, multifaceted discipline that spans a wide range of activities—from requirements elicitation and system design, through implementation and testing, to deployment, maintenance, and evolution. Each phase of the software development lifecycle (SDLC) presents its own unique challenges, and the quality of the final product depends critically on how effectively engineers navigate all of them. Despite decades of research and tooling improvements, many of these activities remain labor-intensive, error-prone, and heavily reliant on individual expertise.

The rapid advancement of artificial intelligence—and generative AI in particular—has opened unprecedented opportunities to augment and support software engineers throughout the entire lifecycle. Tools such as GitHub Copilot and ChatGPT have already demonstrated the potential of LLMs in code generation and debugging, but the scope for AI-driven support extends far beyond writing code. Promising applications include automated requirements analysis, architecture recommendation, intelligent test case generation, code review assistance, predictive maintenance, and project management support. However, most existing tools focus narrowly on individual tasks, and a holistic understanding of how AI can be integrated across the full SDLC remains largely unexplored.

This thesis proposal aims to investigate the design, development, and evaluation of AI-powered tools that support software engineers across one or more phases of the software lifecycle. The work may take different forms depending on the student's interests: it could involve a systematic study of the current state of the art, the design and prototyping of a novel tool targeting a specific lifecycle phase (or bridging multiple phases), or an empirical investigation of how practitioners perceive and adopt AI-based assistants in their daily workflows. Students are encouraged to identify a specific angle or gap and to develop a well-engineered solution or contribution, supported by appropriate evaluation.

The scope of this thesis track is intentionally broad to accommodate diverse research directions. Potential topics include, but are not limited to: AI-assisted requirements engineering, LLM-based design pattern suggestion, intelligent code review and refactoring, AI-driven test generation and mutation testing, automated documentation generation, predictive bug detection, and AI support for DevOps and continuous delivery pipelines. Students are strongly encouraged to contact Stefano Lambiase to discuss their ideas and tailor the thesis focus to their interests and skills.

Potential collaborations with companies in Denmark can be arranged. The professor already has some contacts with software companies in Denmark, but the plan is to search for collaborations with others.

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